PYTHON FOR DATA SCIENCE · LESSON 21 OF 29

Data Visualisation II: Seaborn Statistical Charts and Interactive Plotly

Make statistical charts quickly with Seaborn (distributions, categorical comparisons, relationships, correlation heatmaps, pair plots and facets) using the HR attrition data, and build interactive charts and simple dashboards with Plotly Express and Streamlit.

Python class 16 · Seaborn 0.13 · Plotly 6 · Intermediate · 100 min

By the end of this lesson you can

- Use Seaborn's figure-level and axes-level functions with hue, col and row
- Compare distributions across groups with histplot, kdeplot, boxplot, violinplot and countplot
- Show relationships with scatterplot, regplot, heatmaps of correlations and pairplot
- Build interactive charts with Plotly Express and know when to use Streamlit
- Turn charts into a clear data story

Datasets used

`hr_employees.csv` `telecom_churn.csv`

Seaborn in one idea

Seaborn works directly with **long DataFrames**: you name the columns for `x`, `y`, `hue` (colour), `col` / `row` (small multiples), and it groups, aggregates and adds legends for you.

Python

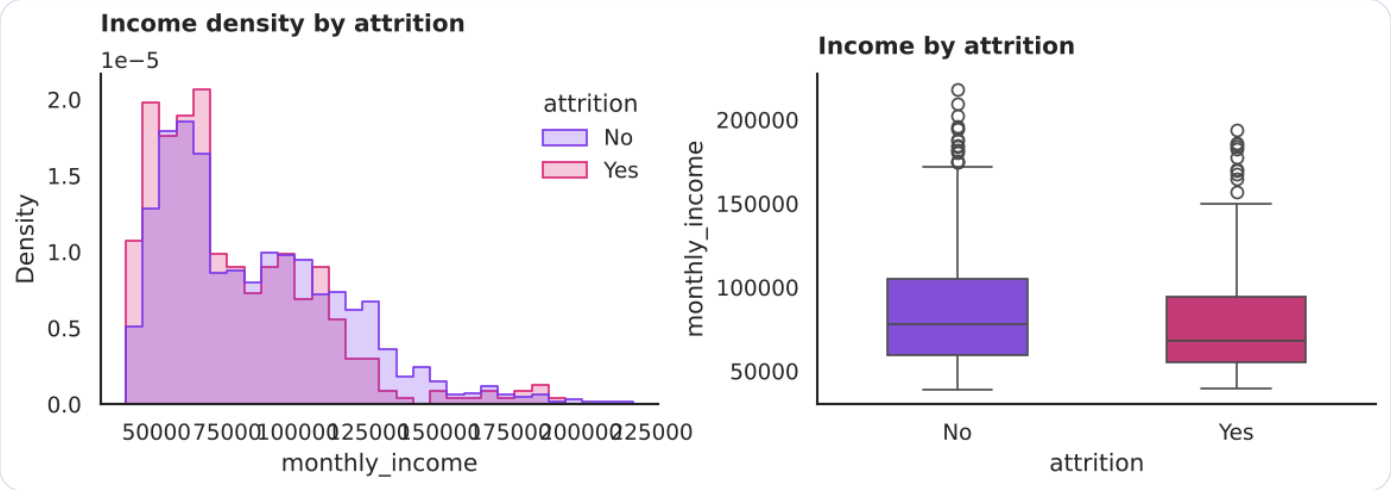
```
import pandas as pd, numpy as np, seaborn as sns, matplotlib.pyplot as plt
hr = pd.read_csv("hr_employees.csv")
hr["left"] = (hr["attrition"] == "Yes").astype(int)
print(hr.shape, f"attrition rate {hr.left.mean():.1%}", sns.__version__)
print(hr.groupby("overtime")["left"].mean().round(3).to_dict(), hr.groupby("department")
["left"].mean().round(3).sort_values().to_dict())
```

OUTPUT

```
(1500, 16) attrition rate 25.9% 0.13.2
{'No': 0.216, 'Yes': 0.367} {'Operations': 0.219, 'Finance': 0.227, 'HR': 0.229,
'IT': 0.243, 'R&D': 0.26, 'Sales': 0.304}
```

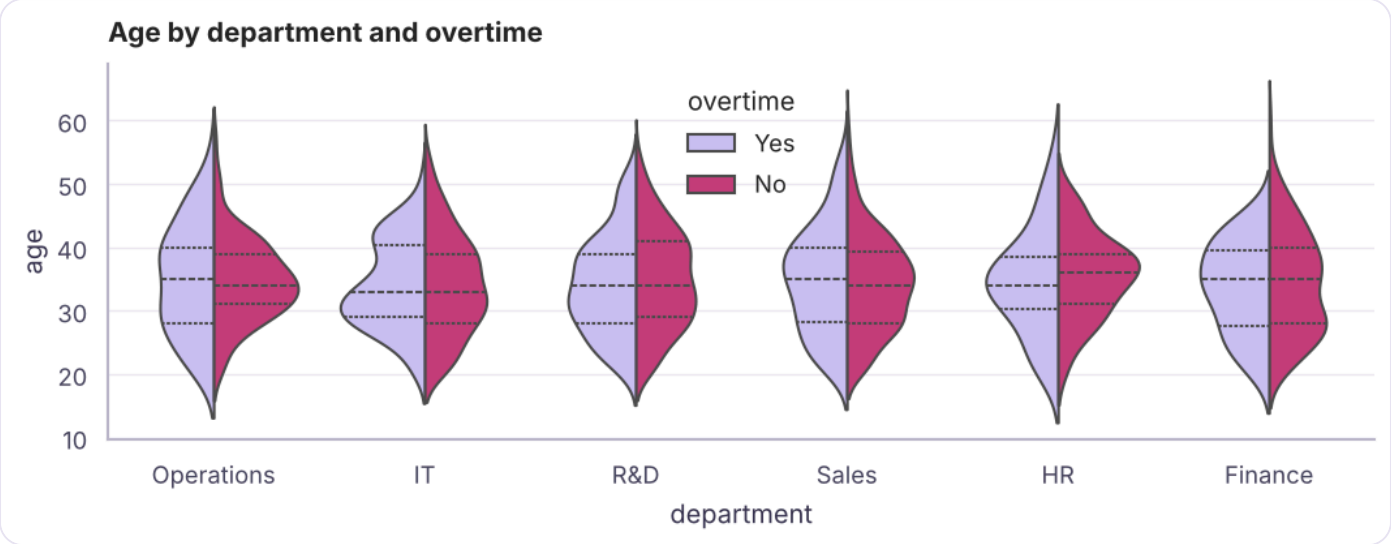
Question	Seaborn function
Distribution of one variable	`histplot`, `kdeplot`, `ecdfplot`
Distribution by group	`boxplot`, `violinplot`, `stripplot`, `boxenplot`
Counts of categories	`countplot`
Average per category (with CI)	`barplot`, `pointplot`
Two numeric variables	`scatterplot`, `regplot`, `lmplot`, `jointplot`
Many variables	`pairplot`, `heatmap` of `df.corr()`
Small multiples	`displot`, `catplot`, `relplot` with `col=` / `row=`

Distributions



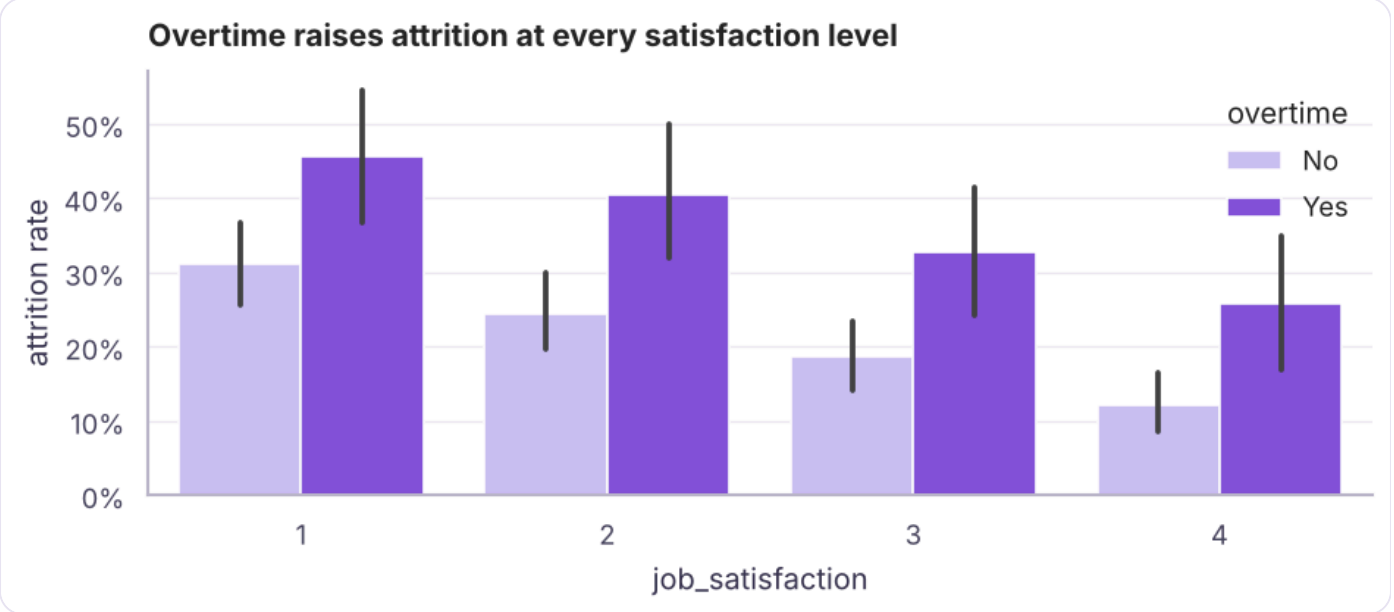
Income distribution for leavers vs stayers: leavers cluster at lower incomes.

`common_norm=False` scales each group separately, so a small group (leavers) is not dwarfed by a big one.



Violin plots combine a box plot with the distribution shape; split by overtime.

Categorical comparisons



Attrition rate by overtime and job satisfaction. Bars show the mean with a 95% confidence interval.

Python

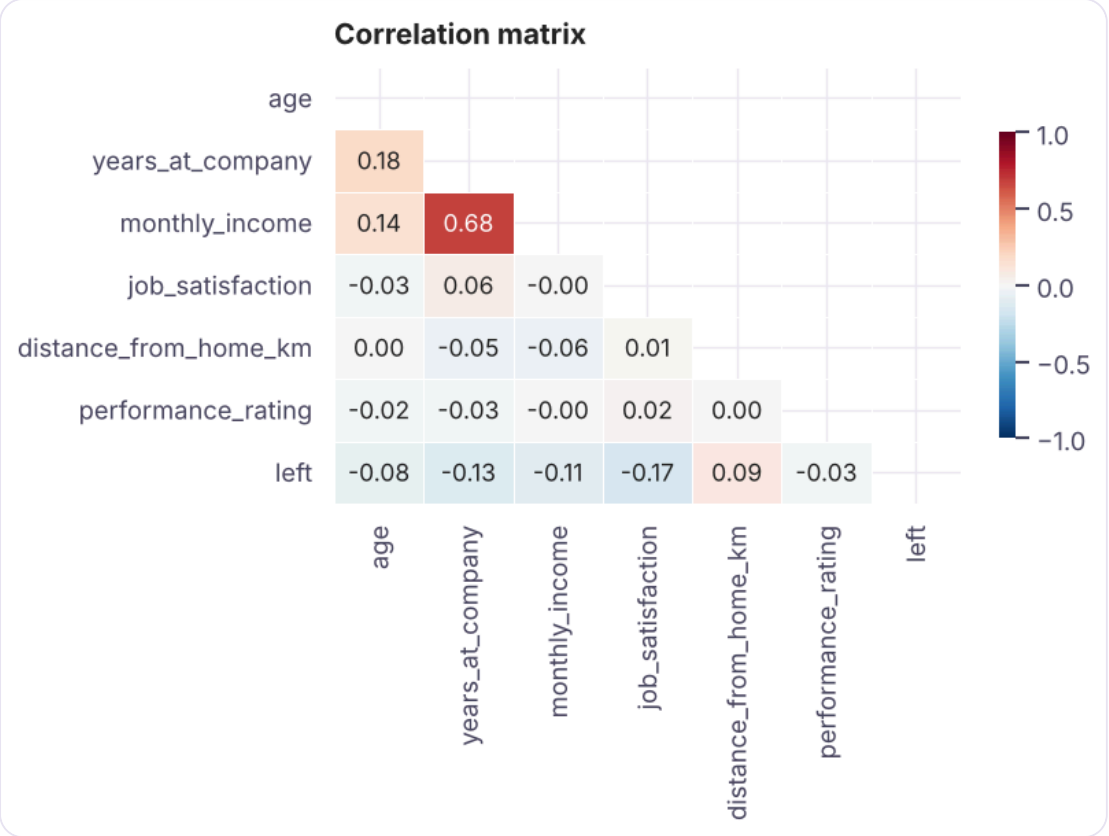
```
print((hr.pivot_table(index="job_satisfaction", columns="overtime", values="left",
aggfunc="mean") * 100).round(1))
```

OUTPUT

```
overtime             No   Yes
job_satisfaction
1                  31.1  45.5
2                  24.4  40.5
```

```
3                 18.7  32.8
4                 12.0  25.8
```

Relationships and correlation



Correlation heatmap of numeric HR variables (Pearson r). Only the lower triangle is shown.

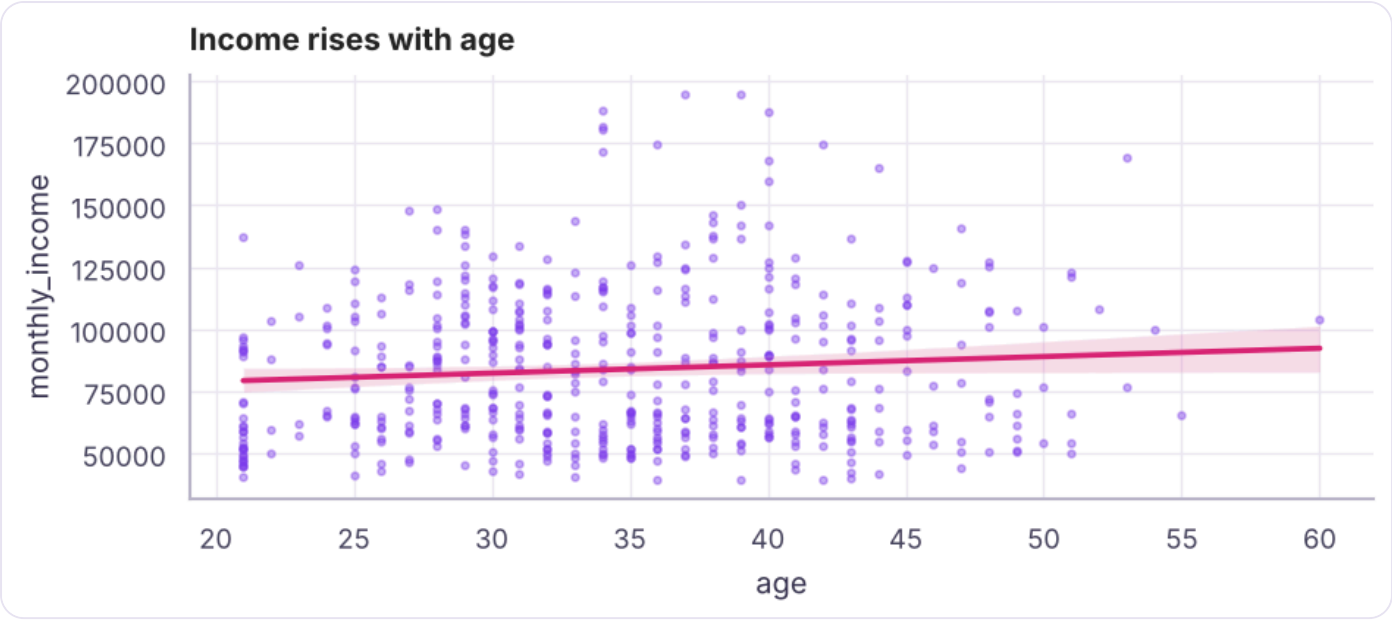
Python

```python
print(corr["left"].drop("left").sort_values())
```

OUTPUT

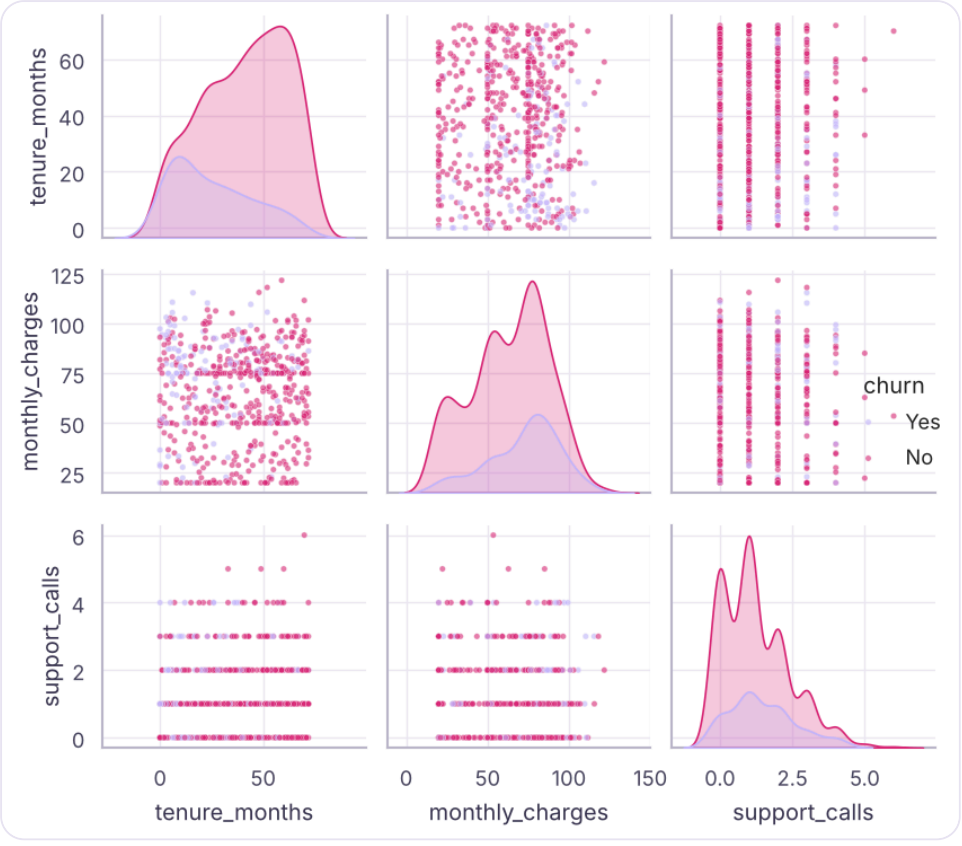
```
job_satisfaction        -0.17
years_at_company        -0.13
monthly_income          -0.11
age                     -0.08
performance_rating      -0.03
distance_from_home_km    0.09
Name: left, dtype: float64
```

Correlation measures **linear** association between −1 and +1. It does not prove causation, and it can miss non-linear patterns: always look at the scatter too.

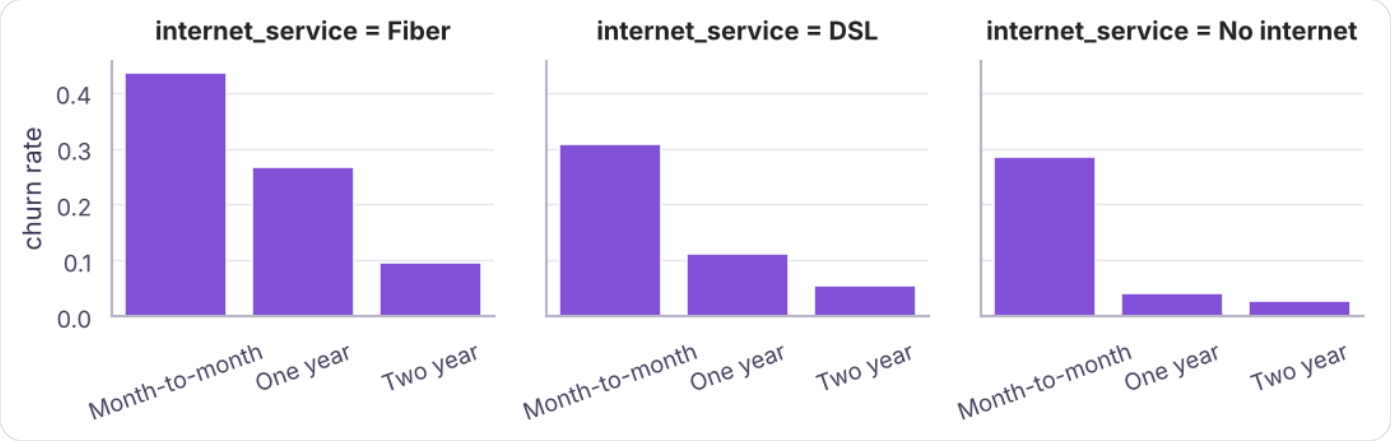


regplot adds a fitted line with a confidence band; lmplot repeats it per group.

Pair plots and facets



A pair plot shows every pairwise scatter and each distribution, coloured by churn (telecom data).



Facets (small multiples) with catplot: churn rate by contract type, split by internet service.

Python

```
print((tc.pivot_table(index="contract", columns="internet_service", values="churned",
aggfunc="mean") * 100).round(1))
```

OUTPUT

```
internet_service   DSL  Fiber  No internet
contract
Month-to-month    30.8   43.5         28.5
One year          11.1   26.7          4.1
Two year           5.6    9.5          2.8
```

Interactive charts with Plotly Express

Plotly makes charts you can hover, zoom and filter in a browser or notebook. The syntax mirrors Seaborn:

Python

```
import plotly.express as px
fig = px.scatter(tc, x="tenure_months", y="monthly_charges", color="churn",
                 hover_data=["contract", "payment_method"], opacity=.6,
                 title="Churn by tenure and charges")
fig.show()                          # interactive in notebooks
fig.write_html("churn.html")        # share as a standalone web page

px.line(monthly_df, x="month", y="sales", color="category")
px.bar(region_df, x="region", y="sales", text_auto=".2s")
px.histogram(hr, x="monthly_income", color="attrition", barmode="overlay")
px.choropleth(...)                  # maps
```

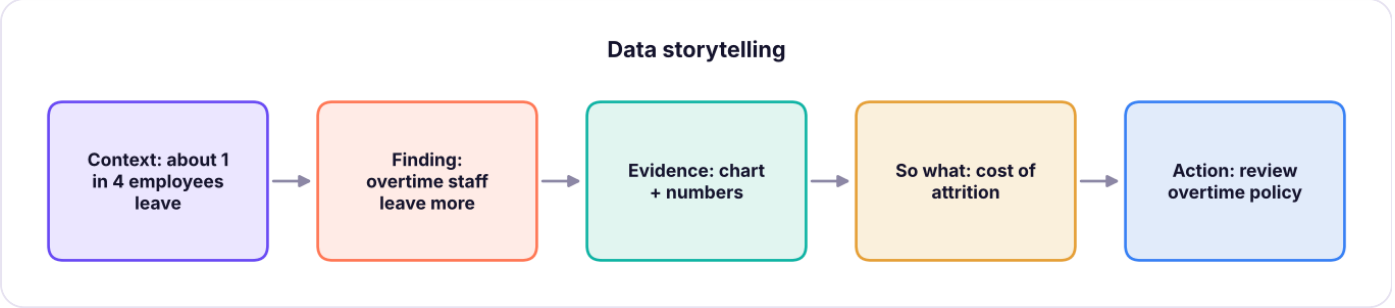
A mini dashboard app with Streamlit

Python

```python
# app.py  →  run with:  streamlit run app.py
import streamlit as st, pandas as pd, plotly.express as px
df = pd.read_csv("telecom_churn.csv")
contract = st.sidebar.multiselect("Contract", df.contract.unique(),
default=list(df.contract.unique()))
d = df[df.contract.isin(contract)]
st.title("Churn dashboard")
st.metric("Churn rate", f"{(d.churn == 'Yes').mean():.1%}")
st.plotly_chart(px.histogram(d, x="tenure_months", color="churn", barmode="overlay"))
```

Streamlit turns a Python script into a web app with filters, which is useful for sharing analyses and ML demos (see the GenAI track for chat apps).

From charts to a data story



The story arc: context, finding, evidence, recommendation.

1. **One message per chart**, written as the title.
2. Show the comparison that proves it (overtime vs not), not every chart you made.
3. Quantify impact (people, ₹, %).
4. End with a recommendation and its owner.

Summary

KEY TAKEAWAYS

- Seaborn plots long DataFrames by column name with `hue`, `col` and `row`; it sits on Matplotlib, so you can still style the axes.
- Distributions: `histplot` (with `common_norm=False`), `kdeplot`, `boxplot`, `violinplot`; categories: `countplot`, `barplot` with confidence intervals.
- Relationships: `scatterplot`, `regplot`, `pairplot`, correlation `heatmap` (correlation ≠ causation).
- Facets (`catplot`, `relplot`, `displot`) create small multiples.
- Plotly Express gives interactive charts; Streamlit turns scripts into dashboards. Tell a story: one message per chart.

Quick quiz

6 questions · tap an answer to check it

1. What does `hue="attrition"` do in Seaborn?

A Changes the theme

B Colours the marks by the attrition category

C Filters to attrition = Yes

D Adds a title

Answer: B. `col=` would create separate panels instead.

2. Which chart shows the mean per category with a confidence interval?

A countplot

B barplot

C histplot

D heatmap

Answer: B. `countplot` shows counts.

3. A correlation of −0.8 means...

A No relationship

B A strong negative linear relationship

C One variable causes the other to fall

D A weak relationship

Answer: B. Correlation does not imply causation.

4. Which function creates small multiples of bar charts?

A barplot

B catplot(kind="bar", col=...)

C heatmap

D pairplot

Answer: B. Figure-level functions accept `col` and `row`.

5. What is the main advantage of Plotly over Matplotlib?

A Faster to print

B Interactive hover, zoom and filtering in the browser

C More chart types for PDFs

D No code needed

Answer: B. Export with `write_html`.

6. Why use `common_norm=False` in a histplot with hue?

A So each group's density is scaled to its own size and small groups stay visible

B To remove the legend

C To use counts

D To normalise to 100

Answer: A. Otherwise the larger group dominates.

EXERCISE

Seaborn practice (telecom_churn.csv)

(1) Histogram of `monthly_charges` by churn. (2) Box plot of `tenure_months` by contract. (3) Churn rate by payment method, sorted. (4) Correlation heatmap of numeric columns plus `churned`. (5) Write three insight titles from what you see.

Show solution (worked answer)

Python

```
sns.histplot(tc, x="monthly_charges", hue="churn", common_norm=False, stat="density")
sns.boxplot(tc, x="contract", y="tenure_months")
order = tc.groupby("payment_method")
["churned"].mean().sort_values(ascending=False).index
sns.barplot(tc, x="payment_method", y="churned", order=order, errorbar=None)
sns.heatmap(tc.select_dtypes("number").corr(), annot=True, fmt=".2f", cmap="RdBu_r",
center=0)
# e.g. "Month-to-month customers churn far more", "Churners have short tenure", "Churn
rises with support calls"
```

INTERVIEW CORNER

Questions you may be asked

Q: How would you visualise the relationship between a categorical and a numeric variable? Box or violin plots per category (distribution), or bar/point plots of the mean with confidence intervals.

Q: How do you show correlations among many numeric variables? A correlation heatmap (masked upper triangle) followed by scatter plots for the interesting pairs; a pair plot for a handful of variables.

Q: How do you present findings to non-technical stakeholders? Lead with the recommendation, show one clear chart per point with an insight title, quantify impact, and keep methods in an appendix.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.